

HINTS AND SOLUTIONS

WORKSHEET - 13

Date: 02-09-2010

1. Key: 1
Hint: Given line is $(2x + 3y - 4) + k(3x - 5y - 13) = 0$
Clearly $(1, -2)$ satisfies the line.
2. Key: 2
Hint: $(k + 1)x + (k + 2)y + 5 = 0$
 $(x + 2y + 5) + k(x + y) = 0$
Solve the equations $x + 2y + 5 = 0$ and $x + y = 0$.
3. Key: 2
Hint: $\frac{3x - 4y + 7}{5} = \pm \frac{12x + 5y - 2}{13}$
 $\Rightarrow 39x - 52y + 91 = -(60x + 25y - 10) \Rightarrow 39x - 52y + 91 = -(60x + 25y - 10)$
One of the bisectors is $11x - 3y + 9 = 0$.
4. Key: 1, 2
Hint: Equation to angle bisectors are $\frac{7x - y + 3}{\sqrt{50}} = \pm \frac{x + y - 3}{\sqrt{2}} \Rightarrow 7x - y + 3 = \pm 5(x + y - 3)$
 $\therefore$ The bisectors are $x - 3y + 6 = 0$ and $3x + y - 3 = 0$
The third side will be parallel to the bisectors.
 $\therefore$ Slopes $\frac{1}{3}$ or -3 .
5. Key: 1
Hint: Angle bisectors are mutually perpendicular
 $\therefore$ The bisector $\perp$ to $x + 2y - 5 = 0$ and passing through $(1, 2)$ is $2x - y = 0$
6. Statement I is true
if a, b, c are in A.P
 $\Rightarrow 2b = a + c \Rightarrow a - 2b + c = 0 \Rightarrow a(1) + b(-2) + c = 0$
 $ax + by + c = 0$ passes through $(1, -2)$
7. Key: 3 For the line $3x + 4y + 3 = 0$, $4x + 3y + 8 = 0$
 $a_1a_2 + b_1b_2 = 12 + 12 = 24 > 0$ $c_1c_2 = 24 > 0$
 $\therefore$ acute angle bisector is ,
 $\frac{3x + 4y + 3}{\sqrt{9 + 16}} = \frac{-(4x + 3y + 8)}{\sqrt{9 + 16}} \Rightarrow 7x + 7y + 11 = 0$
8. Key: 2 For the lines $x - y + 2 = 0$ and $7x + y + 1 = 0$
 $a_1a_2 + b_1b_2 = 7 - 1 = 6 > 0$ $c_1c_2 = 2 > 0$
obtuse angle bisector is, $\frac{x - y + 2}{\sqrt{1 + 1}} = \frac{7x + y + 1}{5\sqrt{2}}$
 $6x + 2y - 1 = 0$

9. Key: 1 As C_1, C_2 have the same signs, the bisector containing origin is

$$\frac{3x - 4y + 2}{5} = \frac{12x - 5y + 2}{13}$$

$$\Rightarrow 39x - 52y + 26 = 60x - 25y + 10 \Rightarrow 21x + 27y - 16 = 0$$

10. Key - q,p,s,r

Hint : (a) $(x - 3y + 9) + k(3x - 2y - 1) = 0$

$$\Rightarrow \text{solve equations } x - 3y + 9 = 0, 3x - 2y - 1 = 0$$

(b) $a(x + y + 1) + b(2x - y + 5) = 0$

$$\text{solve equations } x + y + 1 = 0, 2x - y + 5 = 0$$

(c) Solve equations $2x + 3y + 1 = 0, 3x - 2y - 5 = 0$

(d) Solve equations $x + y - 4 = 0, 2x - y - 2 = 0$

11. Key: 4 Clearly the line $2x - y + 5 = 0$ does not contain $(5, -2)$ so $(x + y + 1) + \lambda(2x - y + 5) = 0$ passes through $(5, -2)$

$$\Rightarrow 4 + \lambda(17) = 0 \Rightarrow \lambda = \frac{-4}{17}$$

$$\therefore \text{The equation of the required line is } 17(x + y + 1) - 4(2x - y + 5) = 0$$

$$\Rightarrow 3x + 7y - 1 = 0.$$

12. Sol: 2,3

For the two lines $24x + 7y - 20 = 0$ and $4x - 3y - 2 = 0$, the angle bisectors are

$$\text{given by } \frac{24x + 7y - 20}{25} = \pm \frac{4x - 3y - 2}{5}$$

Taking positive sign, we get $2x + 11y - 5 = 0$.

13. Key: 1

Hint: $(2x - 3y + 2) + k(5x - 6y - 1) = 0$

Then solve $2x - 3y + 2 = 0$ and $5x - 6y - 1 = 0$

14. Key: 1,3 Sol: It is clear that diagonals of the rhombus will be parallel to the bisector of the given lines and will pass through $(1, 3)$. Equations of bisectors of the given lines

$$\text{are } \frac{x + y - 1}{\sqrt{2}} = \pm \left(\frac{7x - y - 5}{5\sqrt{2}} \right) \text{ or } 2x - 6y = 0, 6x + 2y = 5$$

Therefore, the equations of diagonals are $x - 3y + 8 = 0$ and $3x + y - 6 = 0$. Thus the required vertex will be the point where these lines meet the line $y = 2x$.

Solving these lines we got possible coordinates as $(8/5, 16/5)$ and $(6/5, 12/5)$.

15. Key: 4

Hint: The lines $x + y - 1 = 0$ and $x - y - 1 = 0$ are perpendicular the lines form a right angled triangle

$$\text{The bisector of angle between the lines } \frac{x + y - 1}{\sqrt{2}} = \pm \frac{(x - y - 1)}{\sqrt{2}} \Rightarrow y = 0 \text{ and } x = 1$$

No bisector is $\perp$ r to $x + 3y + 3 = 0$

$\Rightarrow$ No isosceles triangle is formed.

16. Firstly, make the constant terms (c_1, c_2) positive $3x - 4y + 7 = 0$ and $-12x - 5y + 12 = 0 \Rightarrow a_1a_2 + b_1b_2 = (3)(-12) + (-4)(-5) = -36 + 20 = -16$
Hence “-” sign gives the obtuse bisector.

$$\Rightarrow \text{Obtuse bisector is } \frac{(3x - 4y + 7)}{\sqrt{(3)^2 + (-4)^2}} = -\frac{(-12x - 5y + 12)}{\sqrt{(-12)^2 + (-5)^2}}$$

$$\Rightarrow 13(3x - 4y + 7) = -5(-12x - 5y + 2)$$

$$\Rightarrow 21x + 77y - 101 = 0 \text{ is the obtuse angle bisector.}$$

COMPREHENSION TYPE:

Equations of bisector of the angles between the given lines are

$$\frac{4x + 3y - 6}{\sqrt{4^2 + 3^2}} = \pm \frac{5x + 12y + 9}{\sqrt{5^2 + 12^2}} \Rightarrow \frac{4x + 3y - 6}{5} = \pm \frac{5x + 12y + 9}{13}$$

$$\Rightarrow 52x + 39y - 78 = \pm(25x + 60y + 45) \Rightarrow 27x - 21y - 123 = 0, 77x + 99y - 33 = 0$$

$$\Rightarrow 9x - 7y - 41 = 0, 7x + 9y - 3 = 0$$

Let the angle between the line $4x + 3y - 6 = 0$ and bisector $9x - 7y - 41 = 0$

$$\text{be '}\theta\text{'}, \text{ then } \tan\theta = \left| \frac{-\frac{4}{3} - \frac{9}{7}}{1 + \left(\frac{-4}{3}\right)\left(\frac{9}{7}\right)} \right| = \frac{11}{3} > 1$$

17. Key: 1 The bisector of the obtuse angle is $9x - 7y - 41 = 0$
18. Key: 2 The bisector of the obtuse angle is $7x - 9y - 3 = 0$
19. Key: 3 The bisector of the angle containing the origin

$$\frac{-4x - 3y + 6}{\sqrt{(-4)^2 + (-3)^2}} = \frac{5x + 12y + 9}{\sqrt{5^2 + 12^2}} \Rightarrow 7x + 9y - 3 = 0$$

20. Key: 1 For the point (1, 2), $4x + 3y - 6 = 4(1) + 3(2) - 6 > 0$
 $5x + 12y + 9 = 12 \times 2 + 9 > 0$
Hence equation of the bisector of the angle containing the point (1, 2)
is $\frac{4x + 3y - 6}{5} = \frac{5x + 12y + 9}{13} \Rightarrow 9x - 7y - 41 = 0..$

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1. Key: 2.

Sol: A(-2,3). Let the point be P (x,y)

$$PA = 5 \Rightarrow PA^2 = 25 \Rightarrow (x + 2)^2 + (y - 3)^2 = 25 \Rightarrow x^2 + y^2 + 4x - 6y - 12 = 0$$

2. Key: 1.

Sol: Origin O(0,0). Let the point be P (x, y)

$$OP = 5 \Rightarrow OP^2 = 25 \Rightarrow x^2 + y^2 = 25$$

3. Key: 1

Sol: Let the point be P (x,y). Distance of P from x - axis)² + (Distance of P from y -

$$\text{axis})^2 + (\text{Distance of P from y - axis})^2 = 25.$$

$$\Rightarrow (|y|)^2 + (|x|)^2 \Rightarrow x^2 + y^2 = 25$$

4. Key : 2,3

Sol: Let the point be P(x,y)

$$\text{Distance from P to x - axis is } 2 \Rightarrow |x| = 2 \Rightarrow x^2 = 4 \Rightarrow x^2 - 4 = 0$$

5. Key:1

Sol: $A(a,0).B(-a,0)$. Let $P = (x,y)$

$$PA^2 + PB^2 = 2c^2 \Rightarrow (x-a)^2 + y^2 + (x+a)^2 + y^2 = 2c^2 \Rightarrow 2x^2 + 2y^2 + 2a^2 = 2c^2$$

$$\Rightarrow x^2 + y^2 + a^2 - c^2 = 0$$

6. Key: 1

Sol: $A(2,3)B(-3,4)$ Let $P = (x,y)$

$$\text{Area of } \triangle PAB = 8.5 \Rightarrow \frac{1}{2} \begin{vmatrix} x-2 & x+3 \\ y-3 & y-4 \end{vmatrix} = \frac{17}{2}$$

$$\Rightarrow |15 - 5y + 2 - x| = 17 \Rightarrow (x + 5y - 17)^2 = 17^2 \Rightarrow x^2 + 10xy + 25y^2 - 34x - 170y = 0$$

7. Let $P = (x,y)$. Given that area of $\triangle PAB = 12$ sq. unit.

$$\Rightarrow \frac{1}{2} \begin{vmatrix} 2+4 & 3-4 \\ 2-x & 3-y \end{vmatrix} = 12 \Rightarrow \begin{vmatrix} 6 & -2 \\ (2-x) & (3-y) \end{vmatrix} = 24$$

$$\Rightarrow |6(3-y) - (-2)(2-x)| = 24 \Rightarrow |22 - 6y - 2x| = 24 \Rightarrow |-2(x + 3y - 11)| = 24$$

$$\Rightarrow |-(x + 3y - 11)| = 12 \Rightarrow (x + 3y - 11)^2 = 144 \Rightarrow x^2 + 6xy + 9y^2 - 22x - 66y - 23 = 0$$

8. Key:3

Sol: Let $P = (x,y)$, Given that area of $\triangle POB = 2 \times$ area of $\triangle POA$

$$\Rightarrow \frac{1}{2} \begin{vmatrix} 0-0 & 0-6 \\ 0-x & 0-y \end{vmatrix} = 2 \times \frac{1}{2} \begin{vmatrix} 0-4 & 0-0 \\ 0-x & 0-y \end{vmatrix} \Rightarrow \begin{vmatrix} 0-6 \\ -x-y \end{vmatrix} = 2 \begin{vmatrix} -4 & 0 \\ -x & -y \end{vmatrix}$$

$$\Rightarrow |-6x| = 2|4y| \Rightarrow 36x^2 = 64y^2 \Rightarrow 9x^2 = 16y^2 \Rightarrow 9x^2 - 16y^2 = 0$$

9. Key: s,r,q,p

a) Let the point be P(x,y)

$$\text{Distance from P to x - axis is } 5 \Rightarrow |y| = 5 \Rightarrow y^2 = 25 \Rightarrow y^2 - 25 = 0$$

b) Let the point be P(x,y)

Distance from P to x - axis = Distance from P to y - axis

$$\Rightarrow |y| = |x| \Rightarrow y^2 = x^2 \Rightarrow x^2 - y^2 = 0$$

c) $A(2,3) B(-2,5)$. Let the point be P(x,y)

$$PA = PB \Rightarrow PA^2 = PB^2 \Rightarrow (x-2)^2 + (y-3)^2 = (x+2)^2 + (y-5)^2$$

$$\Rightarrow -4x - 6y + 13 = 4x - 10y + 29 \Rightarrow 8x - 4y + 16 = 0 \Rightarrow 2x - y + 4 = 0$$

d) $A(-3, 4) B(3, 4)$. Let the point be (x, y) $PA = PB \Rightarrow PA^2 = PB^2$

$$(x+3)^2 + (y-4)^2 = (x-3)^2 + (y-4)^2$$

$$x^2 + x + 6x = x^2 + 9 - 6x \Rightarrow 12x = 0 \Rightarrow x = 0$$

10. Key: 4

Sol: Let the point be $P(x, y)$

Distance from P to x -axis : Distance from P to y -axis is $= 2 : 3$

$$\Rightarrow |y| : |x| = 2 : 3 \Rightarrow |y| = 2|x| \Rightarrow 4x^2 - 9y^2 = 0$$

11. $A(4, 0)$. Let the point be $P(x, y)$

$$PA = 2 \times \text{Distance from } P \text{ to } x\text{-axis} \Rightarrow PA = 2|y| \Rightarrow PA^2 = 4y^2$$

$$\Rightarrow (x-4)^2 + y^2 = 4y^2 \Rightarrow x^2 - 3y^2 - 8x + 16 = 0$$

12. $(x-4)^2 + (y-3)^2 = 4[(x-5)^2 + (y+3)^2]$ which represents a circle equation.

13. Key: 3

Sol: $A(0, 5)$. Let the point be $P(x, y)$

$$PA = 2|x| \Rightarrow PA^2 = 4x^2 \Rightarrow x^2 + (y-5)^2 = 4x^2 \Rightarrow 3x^2 - y^2 + 10y - 25 = 0$$

14. $A(1, 2)$; origin $O(0, 0)$. Let the point be $P(x, y)$

$$PO = 2PA$$

$$\Rightarrow PO^2 = 4PA^2 \Rightarrow x^2 + y^2 = 4[(x-1)^2 + (y-2)^2] \Rightarrow 3x^2 + 3y^2 - 8x - 16y + 20 = 0$$

15. Key : s, p, q, r

16. $A(5, -4) B(7, 6)$. Let the point be $P(x, y)$

$$PA : PB = 2 : 3 \Rightarrow 3PA = 2PB \Rightarrow 9PA^2 = 4PB^2 \Rightarrow 9[(x-5)^2 + (y+4)^2] = 4[(x-7)^2 + (y-6)^2]$$

$$\Rightarrow 9(x^2 + y^2 - 10x + 8y + 41) = 4(x^2 + y^2 - 14x - 12y + 85) \Rightarrow 5x^2 + 5y^2 - 34x + 120y + 79 = 0$$

17. Key: 1

$$\text{The locus of } P \text{ is } \frac{4(x-2)^2}{8^2 - 4(3)^2} + \frac{4y^2}{8^2} = 1 \Rightarrow \frac{4(x-2)^2}{28} + \frac{y^2}{16} = 1$$

$$\Rightarrow 16(x^2 - 4x + 4) + 7y^2 = 112 \Rightarrow 16x^2 + 7y^2 - 64x - 48 = 0.$$

18. Key: s, r, q, p

19. Key: 1

$$\text{Sol: The locus of } P \text{ is } \frac{4x^2}{c^2} + \frac{4(y-0)^2}{c^2 - 4a^2} = 1 \Rightarrow 4(c^2 - 4a^2)x^2 + 4c^2y^2 = c^2(c^2 - 4a^2)$$

20. Key: 1

$$\text{Sol: The locus of } P \text{ is } \frac{4x^2}{4a^2} + \frac{4(y-0)^2}{4a^2 - 4(ae)^2} = 1 \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{a^2(1-e^2)} = 1$$

21. Key: 2 $\tan \beta = \frac{y}{a-x}, \tan \gamma = \frac{y}{a+x}$

$$\cot(\beta - \gamma) = \cot \frac{\pi}{2} = 0$$

$$\Rightarrow \cot \beta \cot \gamma + 1 = 0 \Rightarrow \left(\frac{a-x}{y} \right) \left(\frac{a+x}{y} \right) = -1$$

$$\Rightarrow a^2 - x^2 = -y^2 \Rightarrow x^2 - y^2 = a^2$$

22. Key: 4 $\tan \beta = \frac{y}{a-x}, \tan \gamma = \frac{y}{a+x}$

$$\tan 2\alpha = \tan(\beta - \gamma) = \frac{\tan \beta - \tan \gamma}{1 + \tan \beta \tan \gamma}$$

$$= \frac{\frac{y}{a-x} - \frac{y}{a+x}}{1 + \left(\frac{y}{a-x} \right) \left(\frac{y}{a+x} \right)} = \frac{y(a+x-a+x)}{a^2 - x^2 + y^2} = \frac{2xy}{a^2 - x^2 + y^2}$$

$$\Rightarrow a^2 - x^2 + y^2 = 2xy \cot 2\alpha \Rightarrow x^2 + 2xy \cot 2\alpha - y^2 = a^2$$

